Step 8: Hedging with power futures (ticket PPA-8)

> **Numbers on this page (update for v1.1).** Sections 1-4.5 were written with the illustrative forward of 85 EUR/MWh (v1.0). The market check led to a decision: since v1.1 the model uses the market forward curve (section 4.6), and since v1.2 the market's seasonal shape and a solar cannibalisation trend (section 4.7); the current numbers are in `reports/step08_hedging.md`. The methods and lessons are unchanged.
>
> **Time needed:** about 4.5 hours. Concepts take 60 minutes, code tour and findings 45 minutes, hands-on, notebook and exercises 2 hours, to-dos 45 minutes. **You will be able to:** explain how German power futures hedge a renewable park, compute a minimum-variance hedge for one or several products, judge a hedge by the risk it removes (not by one year's P&L), size the PPA buyer's hedge, and check a model's forward against the market.

Contents: 0. Scenario · 1. Concepts · 2. Architecture · 3. Code tour · 4. Findings (and 4.6 the v1.1 update, 4.7 the v1.2 update) · 5. Hands-on · 6. Results · 7. Exercises · 8. Quiz · 9. Interview · 10. Real work · 11. To-dos · 12. Glossary · 13. Sources

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0. The scenario

Friday, the treasury desk:

> *"Not every MWh will be sold through a PPA. For the output we keep, how much should we sell forward on EEX, in which product, and what does that do to our budget risk? And is the 85 EUR/MWh in your model anywhere near the market? Ticket PPA-8."*

Step 7 priced the PPA and showed that the buyer can hedge the price level with a baseload forward. Step 8 makes that hedge explicit, for the producer's unsold output and for the PPA buyer, and compares the model's forward with published market quotes.

Ticket PPA-8: acceptance criteria (all met today)

- ☑ German futures products on the delivery calendar: Cal base and peak, quarters, months.
- ☑ Hedge payoffs and minimum-variance hedges for one or several products.
- ☑ Strategies for both parks' merchant output: none, 50% and 100% of average output, minimum-variance with Cal base, quarters, months; CFaR and variance removed.
- ☑ The PPA buyer's minimum-variance band against Step 7's full band.
- ☑ A back-test on 2024-2025 and a market check against published quotes.
- ☑ Report, `hedging.json` for the memo and the dashboard, learning page, notebook, exercises.

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1. Concepts you need today

1.1 What a futures hedge pays

A German power future delivers a constant MW over its period at a fixed price. Selling h MW of a product k at forward F_k pays, at settlement,

```
h · Σ_(t in k) (F_k − P_t)
```

the seller gains when spot ends below the forward. Products (EEX, local time):

Product	Hours
Cal base	every hour of the year (8,760)
Cal peak	Monday to Friday 08:00-20:00, holidays included (3,132 in 2027)
Quarter / month base	every hour of the quarter / month

A park that sells at spot and sells futures receives spot revenue **plus** the hedge payoff. If prices fall, the hedge gains what the spot sales lose.

1.2 The minimum-variance hedge

How much to sell? The amount that makes the hedged revenue as stable as possible:

```
h* = argmin Var(C + h X) = − Cov(C, X) / Var(X)
```

for one product (C = revenue per scenario, X = payoff of selling 1 MW). For several products, the same is a least-squares regression of the revenue on the product payoffs. The classic hedge ratio from finance (Ederington, 1979), applied to scenarios.

Hedge ratio = MW sold / average output in MW. Selling 100% of the average output is the intuitive choice; it is not the optimal one when output is concentrated in cheap hours.

1.3 Why less than 100% for solar

A solar park produces when prices are low. If the price level rises by 10 EUR/MWh, its revenue rises less than 10 EUR/MWh × its energy, because in the arcsinh price model high-price hours move more than low-price hours, and solar produces in the low-price ones. A full baseload band then over-hedges: in an expensive year the hedge loses more than the park gains. The minimum-variance amount is about 90% of the average output for solar, 93-95% for wind.

1.4 Judging a hedge

A hedge is good if it **reduces risk**, not if it **made money** in one year. A hedge sold at a fair price has an expected P&L of zero; it wins in some years and loses in others. Two measures:

- **CFaR reduction:** how much smaller the 95% cash flow at risk is.

- **Variance removed** (hedge effectiveness): 1 − Var(hedged) / Var(unhedged). Accounting rules for hedge accounting look at similar effectiveness tests.

1.5 Model forward and market forward

The model's forward is its expected average price (here the assumed 85 EUR/MWh, by construction of the calibration in Step 5). The market forward is what someone will pay today. If they differ, selling forward has an expected gain or loss: the **forward premium**. Before trading on it, the question is who is wrong. A pricing model normally takes the market forward as its level and uses the simulation only for shape and risk.

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2. What was built: architecture

```
Step 7 scenarios (P, q; analog days from 2025-10-01)        config [hedging]: back-test years,
            |                                                market quotes (product, price,
date, URL)
hedging/products.py   product_masks: Cal base/peak, Q1–Q4, M01–M12 (local time)
hedging/hedge.py      model_forwards · hedge_payoffs · min_variance_hedge
                      hedge_strategies · buyer_hedge · backtest_hedge
            |
analysis/step08_report.py ─► reports/step08_hedging.md + 3 charts        (`ppa-lab hedging`)
                          ─► data/processed/hedging.json ─► memo and dashboard (`ppa-lab
memo`)
```

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3. Code tour

3.1 `hedging/products.py` and `hedging/hedge.py`

`product_masks` turns the delivery hours into boolean masks per product, in local time (the March month has 743 hours). `model_forwards` averages the scenario prices over each product's hours. `hedge_payoffs` builds the matrix X (scenarios × products) of selling 1 MW; `min_variance_hedge` solves the least-squares problem on centred values. `hedge_strategies` compares the standard strategies; `buyer_hedge` sizes the PPA buyer's band; `backtest_hedge` sells Cal base for each year at the previous year's average price.

3.2 `analysis/step08_report.py` and the memo

The report uses the same scenarios as Step 7, writes the tables and charts, and stores the results in `hedging.json`. `ppa-lab memo` reads it: the memo's third recommendation now gives the hedge, and its limits paragraph the market check; the dashboard has a Hedging tab.

3.3 Tests

Test file	Protects against
`test_hedging.py` (6)	wrong product hours (peak window, DST month); forwards that are not averages; a hedge that does not recover a known exposure; strategies in the wrong order; a buyer band larger than needed; a back-test that sells at the same year's price
`test_step08_report.py` (2), `test_config.py` (+1), `test_cli.py` (+1)	the report end to end, the missing-weather message, `[hedging]` validation (back-test years, quote products), `ppa-lab hedging` and `all`
`test_step09_memo.py` (+1), `test_dashboard.py` (+1)	the memo and the dashboard using the hedging results

Planted bugs that the tests catch: the hedge with the wrong sign, the back-test forward from the same year, and the peak window shifted by one hour. The first products test only counted peak hours and missed the shifted window (same count); it now checks the exact hours of a Wednesday and a Saturday.

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4. Five real findings

4.1 Selling about 90% of the average output removes most of the risk

Merchant, 2027	No hedge	Cal base 100%	Cal base minimum-variance
Solar park CFaR95	EUR 1.44 m	EUR 0.35 m	**EUR 0.27 m** (5.6 MW, 89%)
Wind park CFaR95	EUR 2.95 m	EUR 0.88 m	**EUR 0.74 m** (11.7 MW, 93%)

A single Cal base contract does almost all the work: quarters and months reduce the CFaR only a little further (solar 0.25 and 0.23).

4.2 Monthly hedges fit noise

The monthly minimum-variance amounts jump around and are even negative in some months (wind January −4.8 MW, November 26 MW). With 496 scenarios, twelve coefficients mostly fit sampling noise; and in this model one annual level shock moves all months together, so there is little for monthly products to add. Real markets have month-specific shocks (gas, hydro, outages), which is why desks do use them, with more data and constraints (no negative sales).

4.3 The PPA buyer needs a smaller band, so the premium is smaller

Step 7 assumed the buyer sells a band of 100% of the average output. The minimum-variance band is 90% (solar) and 95% (wind), and the premium falls from **5.27 to 4.45 EUR/MWh** (solar) and from 3.39 to 2.92 (wind). The Step 7 premium was conservative by about 0.8 EUR/MWh for solar.

4.4 One year of P&L says nothing

Back-test (Cal base at the previous year's average, 89-93% of P50 output): the hedge earned EUR +0.82 m for the solar park in 2024 (sold at 95.18, spot 78.51) and lost EUR 0.53 m in 2025 (sold at 78.51, spot 89.32). With this naive forward the hedged revenue even moved more between the two years than the merchant revenue (EUR 1.27 m vs 0.09 m), because last year's price moved against this year's. Two years are an illustration; the scenario results measure the risk.

4.5 The model's forward is far below the market's

Published Cal-27 base quotes: **104.24 EUR/MWh** (31 Jul 2026, tacto.ai) and **113.45 EUR/MWh** (28 Aug 2026, power2market); peak 119.48. The model uses the assumed 85 from the preparation guide. At 113.45, with the same capture rate, premium and costs, the solar indicative price would be **46.8 EUR/MWh instead of 33.3**, wind 93.0 instead of 68.4. These are secondary web sources, not EEX settlement files; they are in the configuration with date and URL so they can be replaced. The decision whether to re-anchor the model to the market is recorded in the backlog: it moves every headline number.

4.6 Update (v1.1): the forward is now the market's

The market check led to a decision (backlog PPA-8b): the model's price level is now the market's. When the quotes were checked again, Cal-27 base had risen further, from 113.45 (28 Aug) to **140.21 EUR/MWh** (9 Oct 2026, the year's high, driven by gas and weak French nuclear output), and the curve falls steeply: **Cal-28 104.75**, **Cal-29 87.58**. A ten-year PPA therefore follows the curve (`[ppa.forward_curve]`), with the years after 2029 held flat at the Cal-29 quote (long-dated years trade too little for a quote).

10-year pay-as-produced PPA (2027-2036)	v1.0: 85 flat	v1.1: market curve
Solar: forward × capture rate = fair price 2027	85.4 × 0.48 = 41.1	141.0 × 0.63 = 89.1
Solar: fair price, 10 years	41.1	**61.1**
Solar: buyer's risk premium after the hedge	5.3	8.0
Solar: indicative price, 10 years	33.3	**50.6**
Wind: indicative price, 10 years	68.4	**80.1**

Three lessons:

1. **The curve matters as much as the level.** 2027 alone is at 141, but the discount- and energy-weighted forward over the ten years is 96.7 EUR/MWh: the ten-year price sits much closer to Cal-28 and Cal-29 than to Cal-27. A flat 140 would have overstated it badly.
2. **The capture rate moved with the level: check it.** The arcsinh price model (Step 5) raises all hours together in transformed units, so hours near zero are lifted too: the solar capture rate rises from 0.48 (at 85) to 0.63 (at 141), and the share of solar output at negative prices falls from 25% to 9%. History supports the direction for one step: from 2025 to 2026 (January to September) the level rose from 88 to 108 EUR/MWh, prices in the sunniest 10% of hours rose from 8 to 23 (the model predicts about +16), and the park's capture rate held at 0.51 although solar capacity kept growing. But 141 lies outside the data, so Step 7 now also shows the usual shape-factor view (the

capture rate of the recent level, 104 EUR/MWh, with all prices scaled to the forward): solar indicative **43.1 instead of 50.6**; wind barely moves (78.6 vs 80.1). That gap belongs in the conversation with Risk.
3. **The market's seasonal shape is steeper than the model's.** Q1-27 traded at 179.47 and Q2-27 at 114.04 EUR/MWh (14 Sep 2026), against the model's 141.4 and 124.3: the market prices a winter gas risk that 2024-2026 history does not contain. A quarterly forward curve is the next step (v2).

The hedging lessons hold at the new level: Cal base at 83% (solar) and 91% (wind) of the average output cuts the merchant CFaR95 by 78% and 68%, and the PPA buyer's premium with the minimum-variance band is 4.8 instead of 8.0 EUR/MWh (solar).

4.7 Update (v1.2): the market's seasons and a cannibalisation trend

Two steps towards how a pricing desk works:

1. **Each quarter at its market price.** A desk builds an hourly price forward curve (HPFC) that matches every traded product, not only the year. The market's seasonal shape is much steeper than the model's history-based one: on 14 Sep 2026 Q1-27 traded at 179.47 and Q2-27 at 114.04 EUR/MWh with Cal-27 at 135.80. The scenarios now solve one level per quarter so that each quarter averages its forward: the quotes' shape scaled to today's Cal level (Q1 185.3, Q2 117.7, Q3 124.3, Q4 134.3 EUR/MWh; the year still averages 140.21). Months and hours within a quarter still come from history.
2. **A cannibalisation trend in the base case.** Solar's capture rate fell from 0.76 to 0.51 in 2023-2025 as capacity grew. The base case now lets it fall 1 percentage point a year after 2027 (0.62 → 0.53 by 2036), slower than history because batteries and curtailment at negative prices dampen it; ±1 and −2 pp are the sensitivities.

2027-2036, EUR/MWh	**v1.1**	**v1.2**
Solar: fair price 2027	89.1	86.9
Wind: fair price 2027	127.9	130.7
Solar: indicative price, 10 years	50.6	**45.2**
Solar: at the recent level's capture rate	43.1	38.0
Wind: indicative price, 10 years	80.1	**81.8**

The lesson: **the market's winter premium hurts solar and helps wind.** Solar produces in the cheaper Q2 and Q3, wind in the expensive winter. Item 3 of the plan, availability, turned out to be in the models already (solar losses 10% incl. availability; wind 12%: wake, availability, electrical); adding it again would have counted it twice. Grid curtailment is compensated in Germany (§13a EnWG), so it is a contract question, not a price one.

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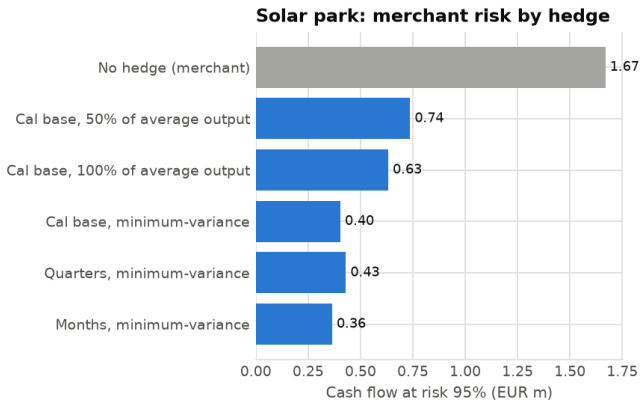
5. Hands-on: run it yourself

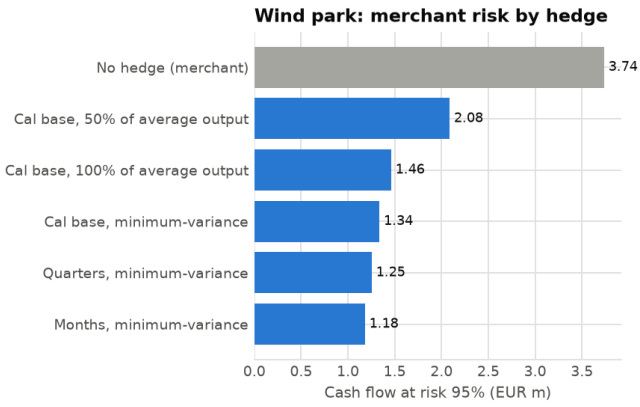
```
cd ".../Encavis quantitative/ppa-lab"
make test                  # all offline tests
make hedging               # strategies, buyer's hedge, back-test, market check (about 7 s)
make memo                  # memo and dashboard results now include the hedge
open reports/step08_hedging.md
make notebook              # Step 8 section: 8.1 to 8.6, then your solutions
```

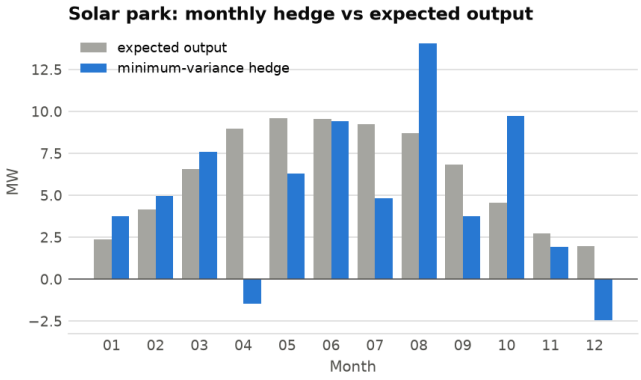
In the notebook: 8.2 draws the CFaR against the hedge size for any product, 8.5 changes the back-test ratio.

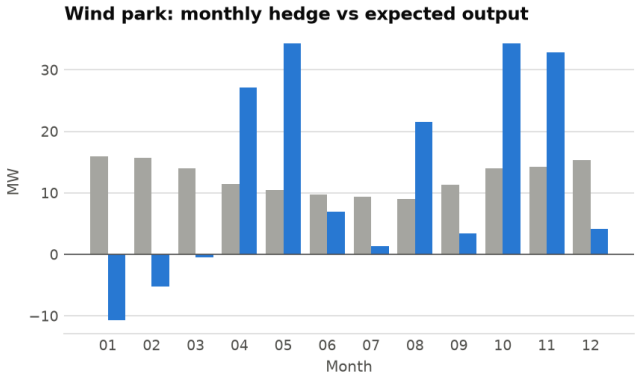
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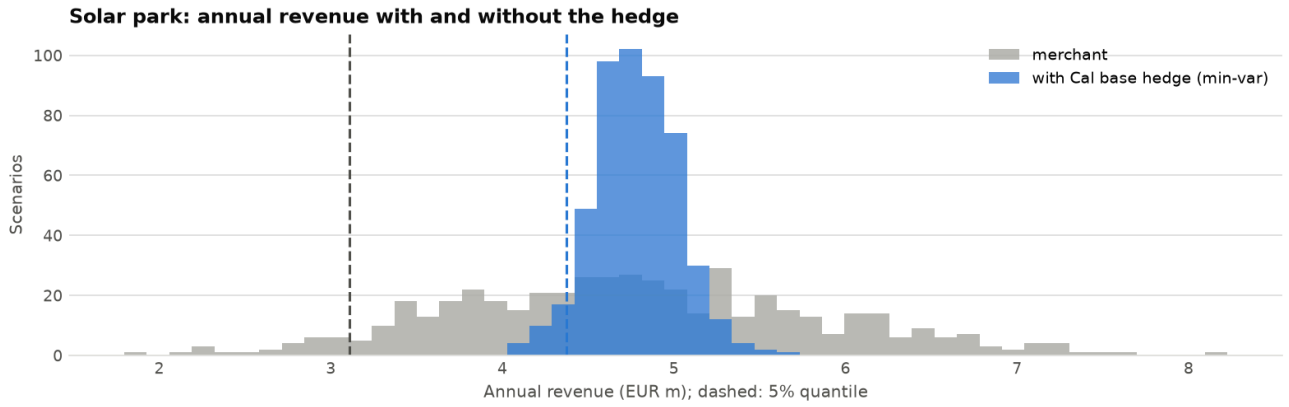
6. Results: what the hedges do











6.1 Five stories

1. **One product does it.** A Cal base hedge of about 90% of the average output removes 97% of the solar park's revenue variance in the model.
2. **100% is too much for solar.** Its output sits in cheap hours; a full band over-hedges.
3. **Finer products need more data.** Monthly hedges from 496 scenarios fit noise.
4. **The PPA buyer's premium shrinks** with an optimal hedge: about 4.5 instead of 5.3 EUR/MWh.
5. **The level is the big unknown.** The market says about 104-113 EUR/MWh for 2027, the model assumes 85: that gap moves the solar price by 13 EUR/MWh, far more than any hedge choice.

6.2 What this means

- For unsold output: sell about 90% of the average output as Cal base; review quarterly products once market data are in.
- For pricing: anchor the level to the market forward (a configuration change, see the backlog), then the simulation provides shape and risk around it.

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7. Exercises

Warm-up (in `exercises/step08_hedging.py`, checked by `.venv/bin/pytest exercises -q`)

1. **What did a hedge earn in a real year?** 5 MW of baseload sold for 2024 and 2025 at the previous year's average.
2. **The minimum-variance hedge** on a toy market.
3. **How good is a hedge?** Variance removed for different sizes.
4. **Written:** "just hedge 100%".

Core

5. **Peak for solar.** Compare a Cal peak hedge with a Cal base hedge for the solar park (notebook 8.2). Why does peak need more MW and do worse?
6. **No negative sales.** Re-solve the monthly hedge with h ≥ 0 (`scipy.optimize.nnls`). How much worse is the CFaR?

7. **The market forward.** Set `forward_eur_mwh = 113.45` in a copy of the configuration and run `make prices scenarios pricing hedging memo`. Which numbers change, and which stay?

Stretch

8. **Volume risk in the hedge.** Split the solar park's residual risk after the hedge into the part that comes from weather years and the part from price paths.
9. **Hedge accounting.** Read about the 80-125% effectiveness test; does the Cal base hedge pass?

▼ **Hints and solution sketches**

1. `forward = prices.loc[str(year - 1)].mean()`, `mw * (forward - prices.loc[str(year)]).sum()`: EUR +0.73 m (2024) and −0.47 m (2025).
2. `-np.cov(cash, payoff)[0, 1] / np.cov(cash, payoff)[1, 1]`: 14.96 MW, 96% of the average output.
3. 0, 0.671 at 100% and 0.672 at the minimum-variance amount: the curve is flat near the optimum, so the exact ratio matters less than avoiding very small or very large hedges.
4. Solar produces in cheap hours, so its revenue moves less than a full band with the level; a 100% hedge loses more than the park gains in an expensive year and the reverse in a cheap one. Judge a hedge by the risk it removes, not by one year's P&L (2024 +0.82, 2025 −0.53). If the market forward is far above the model's expectation, selling forward has an expected gain: either the market knows something (gas, CO2) or the model's level is wrong; check before trading.
5. Peak covers 36% of the hours; the solar park also produces at weekends.
6. Expect a small loss of effectiveness and much more sensible monthly amounts.
7. Prices, capture prices and premia scale with the level; the capture rate and the hedge ratio barely move.

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8. Self-check quiz

1. What does selling 1 MW of Cal base at forward F pay?
2. Write the minimum-variance hedge for one product.
3. What is a hedge ratio here, and why is the optimum below 100% for solar?
4. Why do monthly hedges from 496 scenarios look strange?
5. How do you judge a hedge, and why not by its P&L?
6. Why did the PPA buyer's premium fall in Step 8?
7. What does the back-test with last year's price as forward show, and what does it not show?
8. What is a forward premium?
9. Why should a pricing model take its level from the market forward?
10. Which number in the whole project moves most if the forward changes from 85 to 113?

▼ **Answers**

1. Σ over the year's hours of (F − P_t) per MW.
2. h* = −Cov(C, X)/Var(X), with C the revenue and X the payoff of selling 1 MW.

3. MW sold / average output; solar's revenue moves less than a flat band with the price level, because it produces in cheap hours, so a full band over-hedges.
4. Twelve coefficients estimated from 496 scenarios fit noise, and the model's level shock is annual, so monthly products have little extra to hedge.
5. By the risk it removes (CFaR, variance); a fair hedge has zero expected P&L and will win or lose in a single year by chance.
6. The minimum-variance band (90%) hedges the buyer's position better than a full band.
7. That hedges win and lose in single years; not how good the hedge is, because the forward proxy is not a real market price and two years are too few.
8. The difference between the market forward and the expected spot price.
9. The market price is where hedges and PPAs can actually be traded; the model adds shape and risk.
10. The indicative prices (solar 33.3 → 46.8 EUR/MWh, wind 68.4 → 93.0).

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9. Interview corner

Your 60-second story (STAR):

> *Situation:* treasury asked how much of the unsold output to sell forward, and whether our model's price level matched the market. *Task:* size futures hedges on the joint scenarios and check the forward. *Action:* I modelled EEX base, peak, quarter and month products, computed minimum-variance hedges, compared strategies by cash flow at risk, re-sized the PPA buyer's hedge, back-tested on 2024-2025, and compared the model's forward with published Cal-27 quotes. *Result:* about 90% of average output as Cal base cuts the solar park's merchant CFaR by 81%; monthly hedges only fit noise; the buyer's premium is 0.8 EUR/MWh lower than assumed; and the market's Cal-27 is 104-113 against our 85, which I flagged as the biggest open assumption.

Likely questions and strong answers:

- "*How do you hedge a solar park?*" Sell baseload (or a profile-matched mix) for about 90% of the expected average output, sized by minimum variance on joint scenarios; review with real forward quotes and keep some volume unhedged for weather risk.
- "*Why not hedge 100%?*" Volume and shape: you would over-hedge in years when output is low or falls in cheap hours.
- "*Your hedge lost money in 2025. Was it a bad hedge?*" No: it was sold at a fair price and reduced risk; one year's P&L is not a measure of a hedge.
- "*Your forward is 85, the market says 113. What now?*" Re-anchor the level to the market and keep the simulation for shape and risk; the difference is the most important input to check.

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10. How this maps to real work at Encavis

In this project	In a company
Model forwards from scenarios	EEX settlement prices and a price forward curve (HPFC) from a vendor or the desk
Minimum-variance hedges on 496 scenarios	Hedge strategies with limits, liquidity and credit lines; rolling hedges over several years
Cal base, quarters, months	Cal, quarter, month and week products; peak and off-peak; profile products
Back-test with a naive forward	Back-tests and P&L attribution with real trade prices
Market check with secondary quotes	Daily market data feeds and model-vs-market reconciliation

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11. Your to-dos for Step 8

- □ Read sections 1 and 4. By hand: 5 MW sold at 100 EUR/MWh for 8,760 hours, spot averages 90: what is the hedge P&L? (EUR 0.44 m.)
- □ In the notebook, 8.2: find the hedge size where the solar park's CFaR is lowest.
- □ Exercises 1-4 in `exercises/step08_hedging.py`; check them in "My solutions: Step 8".
- ☑ Decide whether to re-anchor the forward to the market: done in v1.1 (section 4.6).
- □ **Commit Step 8** (commands below).
- □ **German vocabulary of this step:** die Absicherung, das Termingeschäft, der Terminkontrakt (Future), das Grundlastprodukt (Base), das Spitzenlastprodukt (Peak), die Absicherungsquote (hedge ratio), die Varianz, die Wirksamkeit (effectiveness), der Terminaufschlag (forward premium), die Glattstellung, die Lieferperiode.

Commit (check `git status` first: no `.env`, no `M .env.example`):

```
cd ".../Encavis quantitative/ppa-lab"
git status --short
git add src tests exercises config reports docs notebooks app METHODOLOGY.md README.md
CHANGELOG.md Makefile pyproject.toml
git status --short
git commit -m "feat(hedging): futures hedges, buyer's hedge, back-test and market check (PPA-
8)"
git push
gh run watch
```

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12. Glossary

Term	Meaning
Future / forward	Contract to deliver power over a future period at a fixed price
Base / peak	All hours / Monday-Friday 08:00-20:00
Hedge payoff	What a sold contract pays: Σ (F − spot) per MW
Minimum-variance hedge	The hedge amount that makes hedged revenue as stable as possible
Hedge ratio	MW sold / average output
Hedge effectiveness	Share of the revenue variance removed by the hedge
Over-hedging	Selling more than the exposure; the hedge then adds risk
Forward premium	Market forward minus expected spot
Settlement price	The exchange's official daily price of a contract

13. Sources

- Ederington, L. H. (1979): The hedging performance of the new futures markets. *Journal of Finance* 34(1), 157-170 (minimum-variance hedge ratio).
- Hull, J. C. (2022): *Options, Futures, and Other Derivatives*, 11th ed. Pearson (hedging with futures).
- EEX: German power futures contract specifications (base, peak, delivery periods).
- Market quotes: tacto.ai, electricity price page (Cal-27 base 104.24 EUR/MWh, 31 Jul 2026); power2market, markets page (Cal-27 base 113.45, peak 119.48 EUR/MWh, 28 Aug 2026). Secondary sources; replace with EEX settlements.
- Encavis preparation guide: hedging with baseload futures, minimum-variance hedge ratio.